

Agenda

1.) Announcements

- 12.3 and 12.4 Homework due Wednesday
- Module Quiz 24 and HW 12.5 are "due" on Friday (they're both ungraded)
- Final Exam is Monday May 8 at 12:25 pm

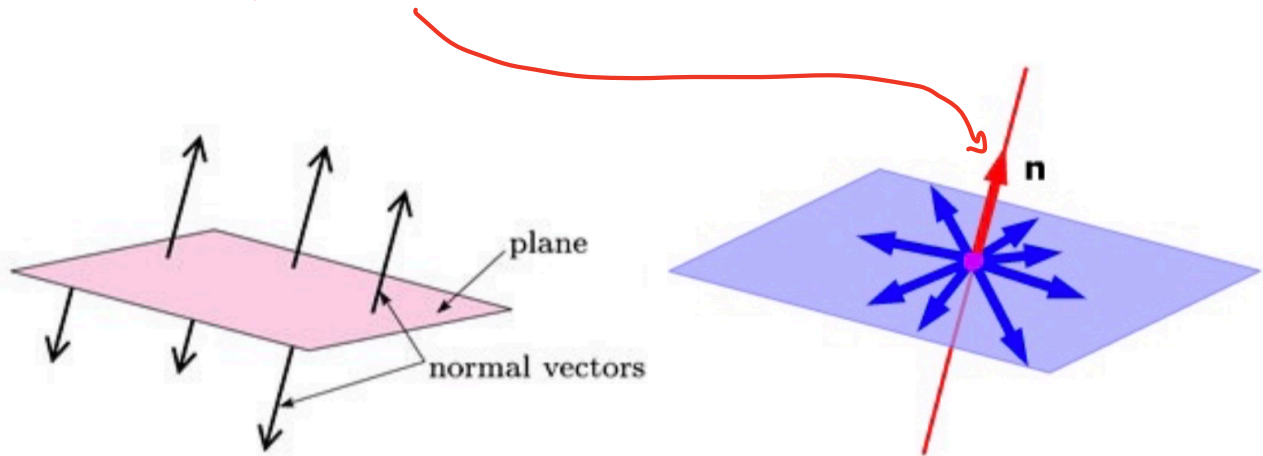
Discussion Section	TA Name(s)	Exam Room Assignment
321, 323, 328, 329, 330, 332	Anuk, Hongjing, and Kevin	Chemistry S429
322, 324, 325, 326, 327	Logan, Jiaming, and Jiaqi	Grainger 1100

(exam info page:
go.wisc.edu/1xxzfx)

2. Section ~~12.4~~ 12.5

Planes

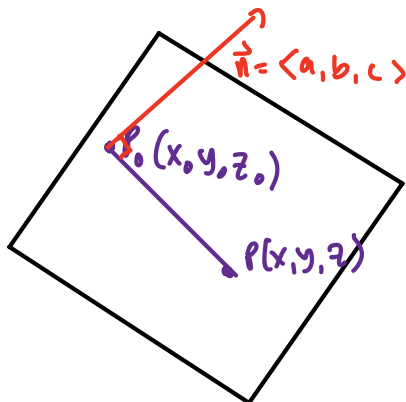
All vectors in a plane are orthogonal to the same direction, called the normal vector to the plane.



If $\vec{v}$ is in the plane, with normal vector $\vec{n}$, then $\vec{v} \cdot \vec{n} = 0$

The plane normal to $\vec{n} = \langle a, b, c \rangle$ containing point $P_0(x_0, y_0, z_0)$ has equation: $\vec{n} \cdot \vec{P_0P} = 0$ $P(x, y, z)$

defines
eqn. of a
plane



$$\langle a, b, c \rangle \cdot \langle x - x_0, y - y_0, z - z_0 \rangle = 0$$

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

$$ax + by + cz = d$$

$$(d = ax_0 + by_0 + cz_0)$$

Activity: Find the equation of the plane that has normal vector $\langle -6, 5, 7 \rangle$ through the point $(1, 2, 3)$.
need: normal vector, point
 $\vec{n} = \langle a, b, c \rangle$
 $P_0(x_0, y_0, z_0)$

$$-6(x-1) + 5(y-2) + 7(z-3) = 0$$

Example: Sketch the plane $2x + 3y + 6z = 6$ in the first octant of $\mathbb{R}^3$.
note: $\vec{n} = \langle 2, 3, 6 \rangle$

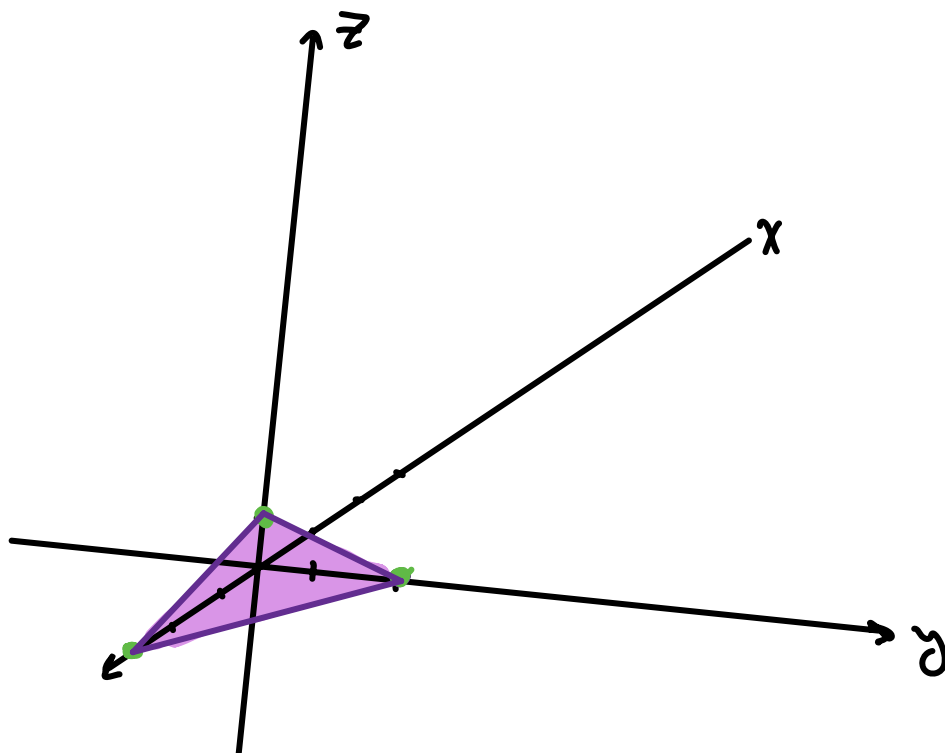
Step 1: Find 3 intercepts

(i) x-int: $(y=0, z=0) \quad 2x=6 \rightarrow x_i = (3, 0, 0)$

(ii) y-int: $y_i = (0, 2, 0)$

(iii) z-int: $z_i = (0, 0, 1)$

Step 2: Plot points and draw the triangle they form.



Three non-collinear points define exactly one plane in $\mathbb{R}^3$
not all on same line

Example: Find the equation of the plane through the points
 $P(0, 1, 2)$, $Q(-1, 3, 0)$, and $R(5, 1, 2)$.

Need normal vector:
find cross product

Let $\vec{v}$ and $\vec{w}$ be two vectors in the plane

$\vec{v} \times \vec{w}$ is a normal vector.

$$\text{Let } \vec{v} = \vec{PQ} = \langle -1, 2, -2 \rangle$$

$$\vec{w} = \vec{PR} = \langle 5, 0, 0 \rangle$$

$$\text{let } \vec{n} = \vec{v} \times \vec{w}$$

$$\begin{aligned} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -1 & 2 & -2 \\ 5 & 0 & 0 \end{vmatrix} = -10\vec{j} - 10\vec{k} \\ &= \langle 0, -10, -10 \rangle \end{aligned}$$

$$\vec{n} = \langle a, b, c \rangle = \langle 0, -10, -10 \rangle$$

$$r_1 = \sqrt{10} \approx 3.1622776601683793319988933444828$$

$$P_0(x_0, y_0, z_0) = (0, 1, 2)$$

$$a(x - x_0) + b(y - y_0) + c(z - z_0)$$

$$0(x - 0) + -10(y - 1) + -10(z - 2) = 0$$

$$-10(y - 1) - 10(z - 2) = 0$$